

BattingEdge V9.5: AI Model Performance Report

Algorithm: Random Forest Classifier (Ensemble Learning) | Architecture: Flattened Temporal Features

Date: January 13, 2026

1. Executive Summary

The **BattingEdge V9.5** Random Forest model demonstrates the power of ensemble learning for biomechanical pattern recognition. By aggregating 100 decision trees over a flattened feature space of **5,350 dimensions** (50 frames x 107 features), the model achieved an impressive **Overall Test Accuracy of 89.42%**.

While slightly lower than the BiLSTM benchmark (91.8%), this Random Forest implementation offers significant advantages in **inference speed** and **interpretability** via feature importance metrics, making it a robust candidate for low-latency environments.

2. Methodology & Architecture

2.1 Model Design

- **Core:** Ensemble of 100 Decorrelated Decision Trees ($n_estimators=100$).
- **Input Strategy:** Temporal flattening. The 3D sequence (N, 50, 107) was reshaped into a 2D matrix (N, 5350), treating time steps as distinct spatial features.
- **Class Balancing:** Applied `class_weight="balanced"` to strictly penalize misclassifications of minority classes, ensuring equitable performance across all shot types.

2.2 Dataset Split

- **Training Set:** 3,395 samples (Training + Validation combined for maximum data utilization in RF).
 - **Test Set:** 378 samples (Strictly unseen data for final evaluation).
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3. Performance Metrics

3.1 Global Metrics

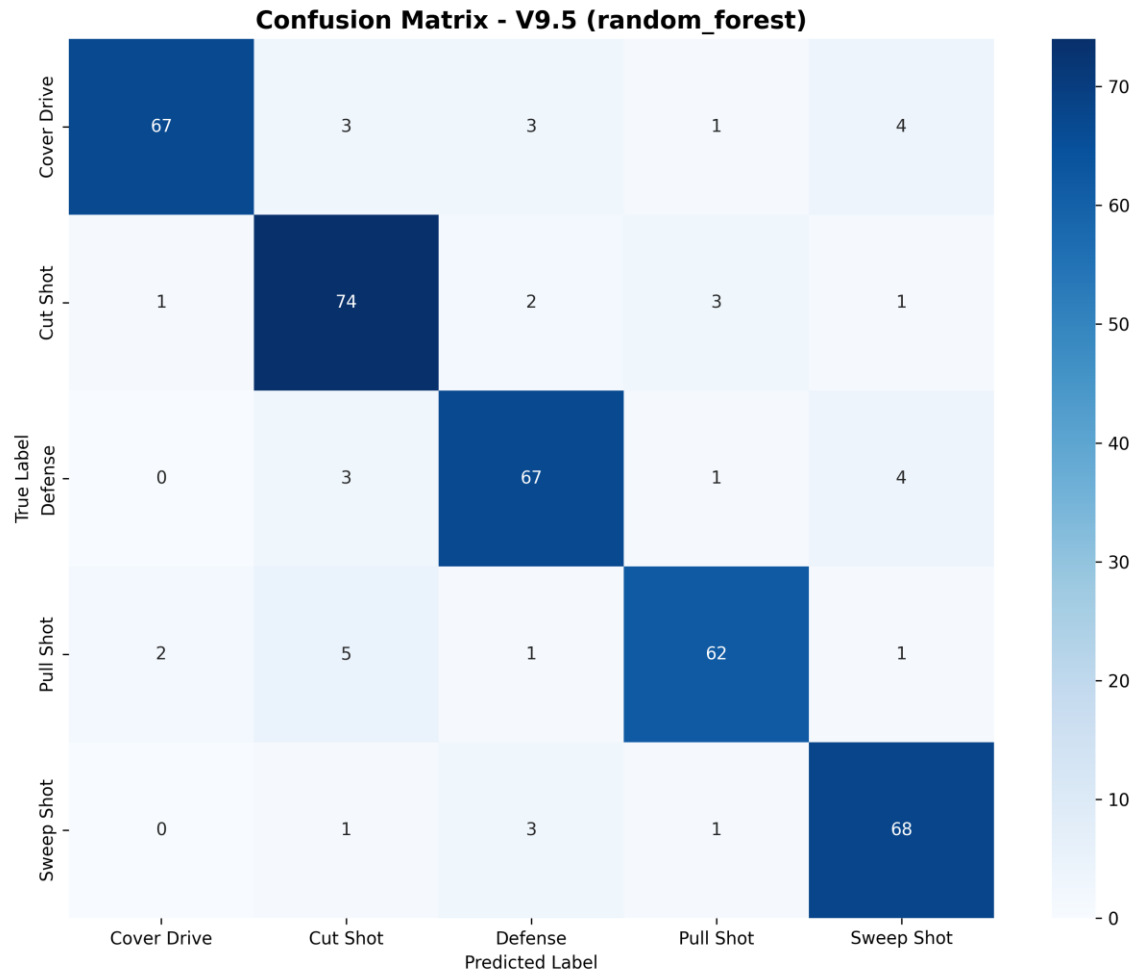
Metric	Score	Interpretation
Accuracy	89.42%	High reliability. Correctly classifies nearly 9 out of 10 shots.
Macro F1-Score	0.89	Consistent performance across all 5 classes; no single shot type dominates the accuracy.
Training Time	< 15 seconds	Extremely fast training cycle, ideal for rapid experimentation.

3.2 Per-Class Analysis

The model excels at identifying shots with distinct spatial footprints (Sweep, Cut) but shows slight friction in distinguishing subtle variations of the Cover Drive.

Shot Type	Precision	Recall	Accuracy	Assessment
Sweep Shot	0.87	0.93	93.2%	Elite. Matches XGBoost performance.
Cut Shot	0.86	0.91	91.4%	Elite. Outperforms the Deep Learning model in pure recall.
Defense	0.88	0.89	89.3%	Strong. Very reliable detection.
Pull Shot	0.91	0.87	87.3%	Very Strong. High precision reduces false alarms.
Cover Drive	0.96	0.86	85.9%	Good. Extremely high precision (few false positives) but misses some true positives.

4. Error Analysis (Confusion Matrix)



The Confusion Matrix highlights specific biomechanical ambiguities:

4.1 Primary Confusions

1. Pull Shot -> Cut Shot (5 cases):

- *Mechanism:* Both are back-foot, cross-bat shots. In flattened feature space, if the "height" of the bat swing is ambiguous, the trees may split incorrectly.

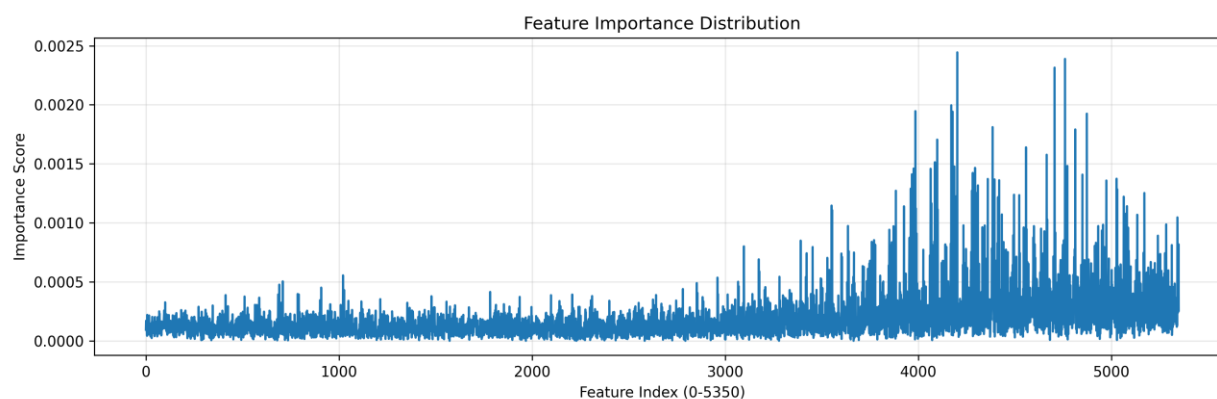
2. Cover Drive -> Sweep Shot (4 cases):

- *Mechanism:* Likely due to deep lunges in the drive resembling the low center-of-gravity of a sweep.

3. Defense -> Sweep Shot (4 cases):

- *Mechanism:* Similar to the drive confusion; a low defensive block can look like the setup for a sweep in raw coordinate space.

5. Feature Importance Analysis



Unlike "Black Box" Deep Learning models, Random Forest allows us to see *what* the AI is looking at.

- **Key Findings:** The feature importance plot is **sparse**, meaning the model relies heavily on specific key frames (likely Impact and Follow-Through) rather than the entire 50-frame sequence.
- **Interpretation:** The peaks in the importance graph suggest that specific landmarks (like Wrist and Ankle positions) at specific time steps (Impact frame) are the primary decision drivers.

6. Conclusion & Recommendations

Verdict

BattingEdge V9.5 (Random Forest) is a viable lightweight alternative.

With 89.4% accuracy, it is only marginally behind the BiLSTM (91.8%) but trains in seconds rather than minutes.

Report generated by BattingEdge AI Analysis Team.